Value Iteration and Q Learning

7 Post

CS 3244
Machine Learning

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Recap and Forecast

Understand how validation and regularization complement to address overfitting:

- **Regularization:** a means of restraining the model.
  - L^p norm forms of regularization.

$J_{test}(h) = J_{train}(h) +$ overfit penalty

- **Validation**, as a reality check by peeking (at the bottom line).
  - Cross Validation variants: k-fold or leave one out.

Learning Outcomes for this week:

- Understand the basic terminology of RL: Policy, reward, action, observation and (environment, agent) model.
- Relate how a **Markov Decision Process** (MDP) embodies RL concepts.
- Understand the **Bellman equations** as a recursive form for calculating future rewards and state values.
- Solve planning and RL problems using the **Value Iteration** and **Q Learning** algorithms.
- Differentiate Value Iteration for planning from Q Learning for RL by their modeling assumptions

Dissecting the MDP

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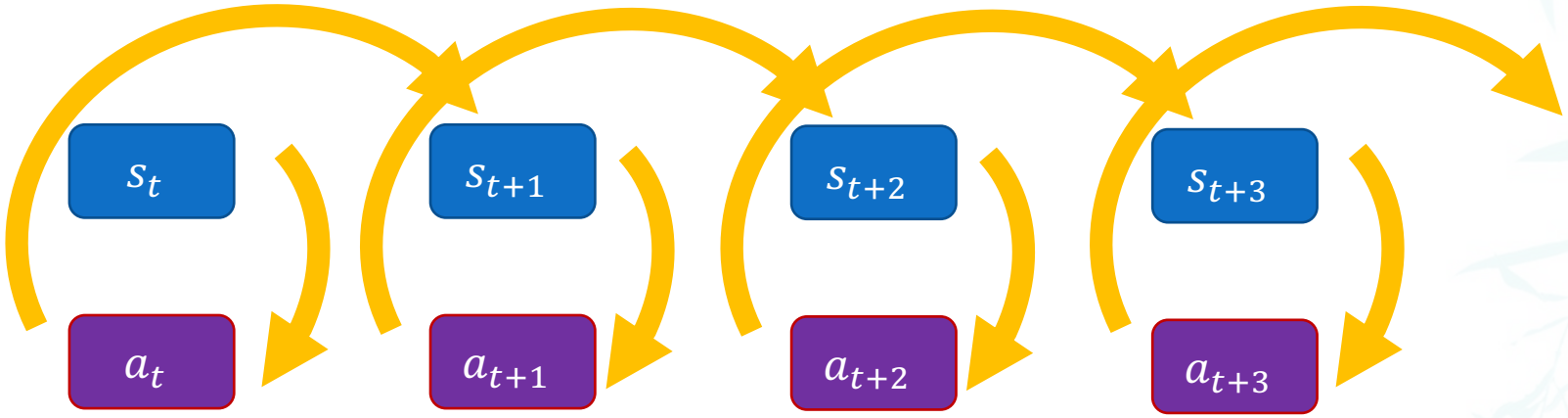
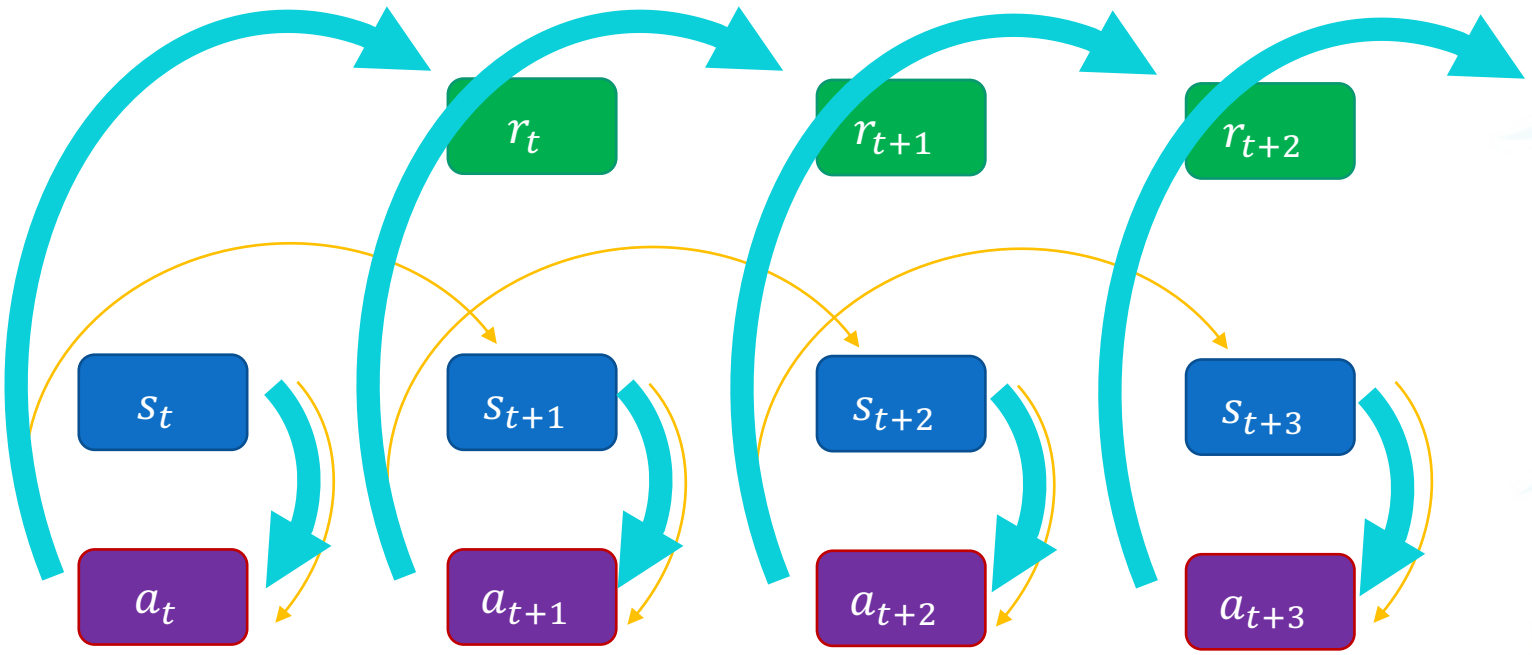
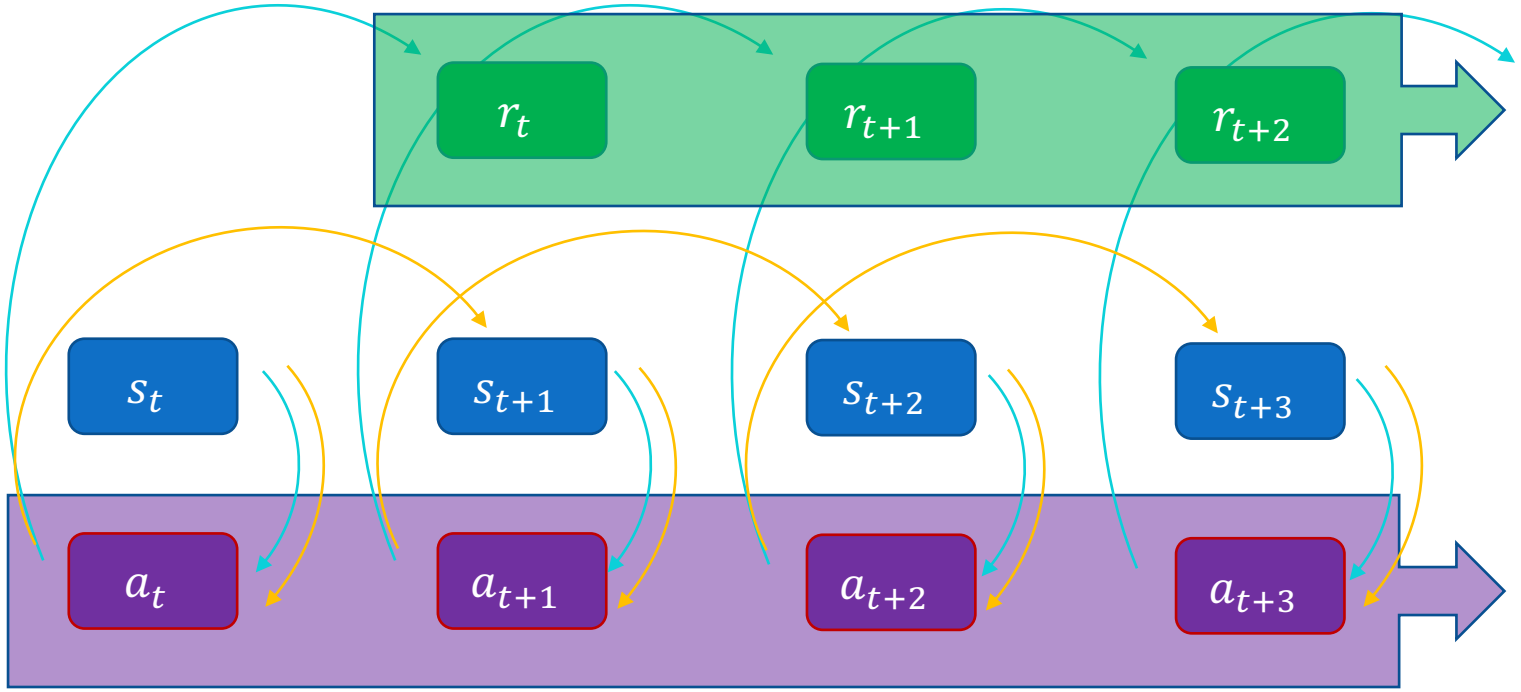
MDP Iterative Formulation



Reward – Iterative formulation



Agent's Actions and Value Estimation



Value function computed based on rewards

Actions chosen based on policy

Return (aka Utility)

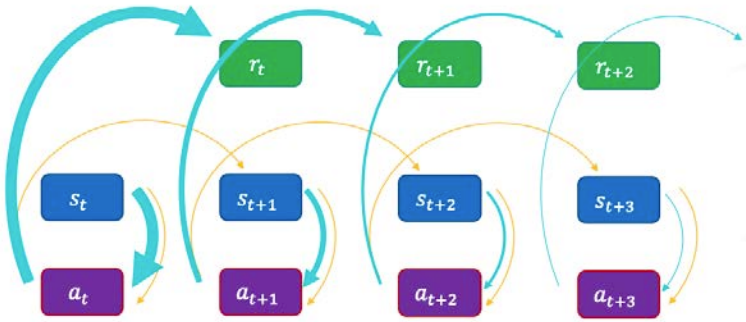
The return g_t is the total discounted reward from time step t.

$$g_t = r_{t+1} + \gamma r_{t+2} + \cdots = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$$

- The discount $\gamma \in [0, 1]$ is the present value of future rewards.
- The value of receiving reward r after $k + 1$ time-steps is $\gamma^k r$.
- This values immediate reward above delayed reward.
  - γ close to 0 leads to "near-sighted" evaluation
  - γ close to 1 leads to "far-sighted" evaluation

Why Discount?

Most Markov reward and decision processes are discounted. Why?



- Mathematically convenient to discount rewards
- Avoids infinite returns from cycles
- Uncertainty about the future
- Animal/human behaviour shows preference for immediate reward
- If the reward is financial, immediate rewards may earn more interest than delayed rewards
- It is sometimes possible to use *undiscounted* Markov reward processes (i.e. $\gamma = 1$), if all sequences terminate.

Value Function

The value function $v(s)$ gives the long-term value of state s.

The state value function $v(s)$ of an MRP is the expected return starting from state s:

$$v(s) = \mathbb{E}(g_t | s_t = s)$$

Policies

A policy π is a distribution over actions given states,

$$\pi(a|s) = P(a_t = a|s_t = s)$$

- A policy fully defines the behavior of an agent
- MDP policies depend on the current state (not the history)
  i.e. Policies are stationary (time-independent):

$$a_t \sim \pi(\cdot\,|s_t), \forall t > 0$$

Value Functions of an MDP

- The *state–value function* $v_\pi(s)$ of an MDP is the expected return starting from state s, and then following policy π:

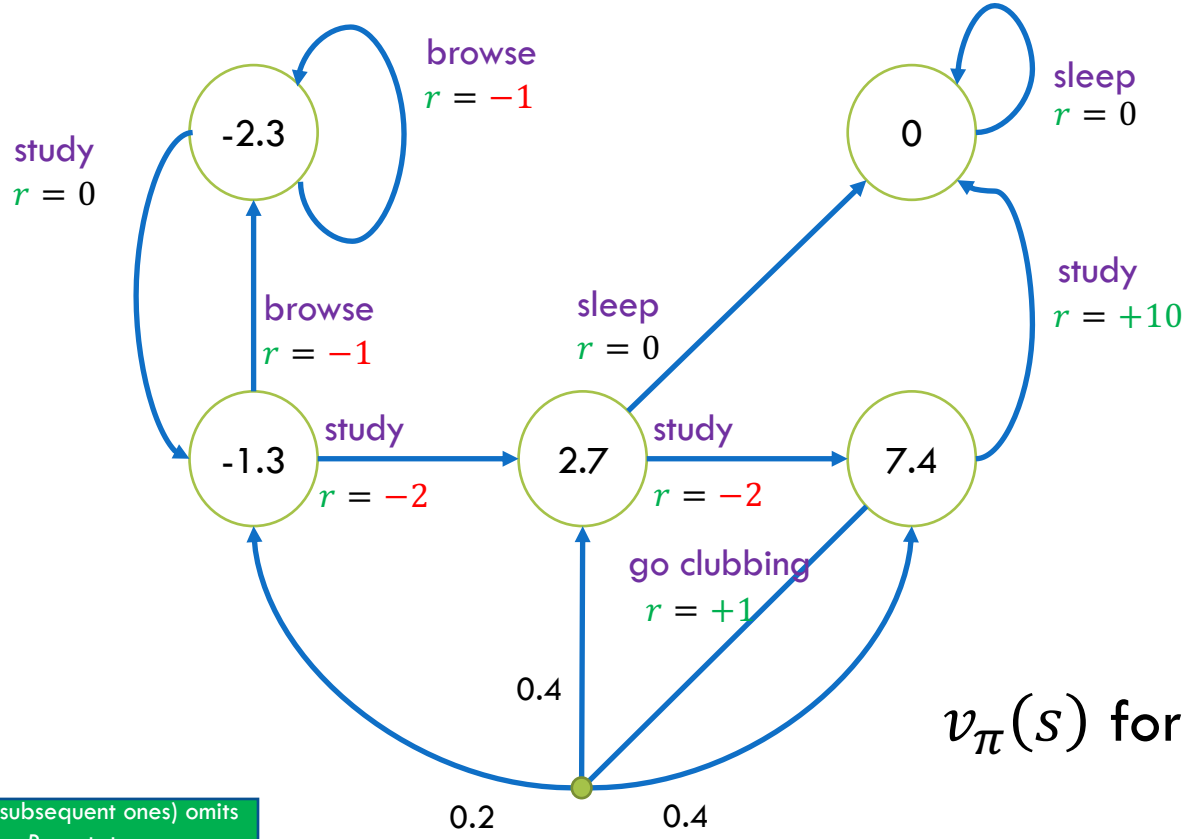
$$v_\pi(s) = \mathbb{E}_\pi(g_t \mid s_t = s)$$

- The *action–value function* $q_\pi(s, a)$ is the expected return starting from state s, taking action a, and then following policy π:

'Q' for Quality

$$q_\pi(s, a) = \mathbb{E}_\pi(g_t \mid s_t = s, a_t = a)$$

Example State–Value Function



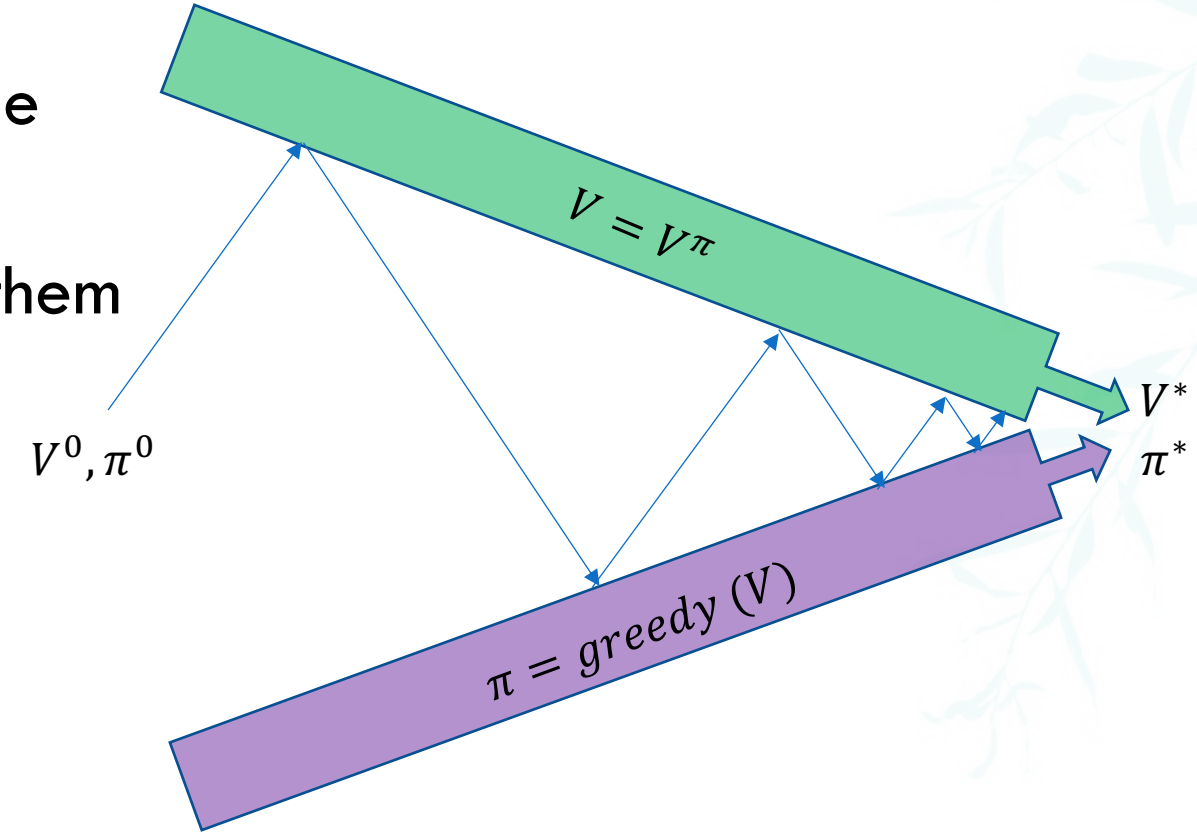
$v_\pi(s)$ for $\pi(a|s) = 0.5, \gamma = 1$

This diagram (and subsequent ones) omits the earlier *Pass* state

Generalised Policy Iteration

Start with a random value function and policy

And iteratively improve them



Bellman Expectation

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Bellman Expectation Equation

The state–value function can be decomposed into immediate reward plus discounted value of successor state:

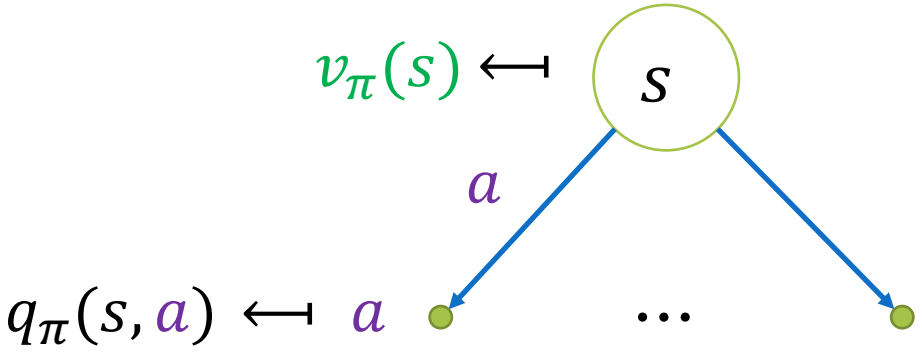
$$v_\pi(s) = \mathbb{E}_\pi\left(\mathcal{R}_{t+1} + \gamma v_\pi(s_{t+1}) | s_t = s\right)$$

The action–value function can similarly be decomposed:

$$q_\pi(s, a) = \mathbb{E}_\pi\left(\mathcal{R}_{t+1} + \gamma q_\pi(s_{t+1}, a_{t+1}) | s_t = s, a_t = a\right)$$

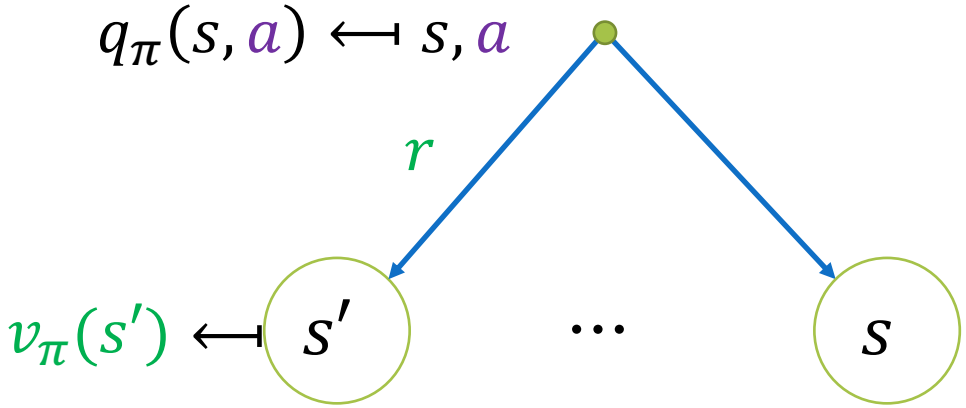
Bellman Expectation Equation for v_π

Bellman Backup (Diagram): Calculate the value function by recursing through one step.



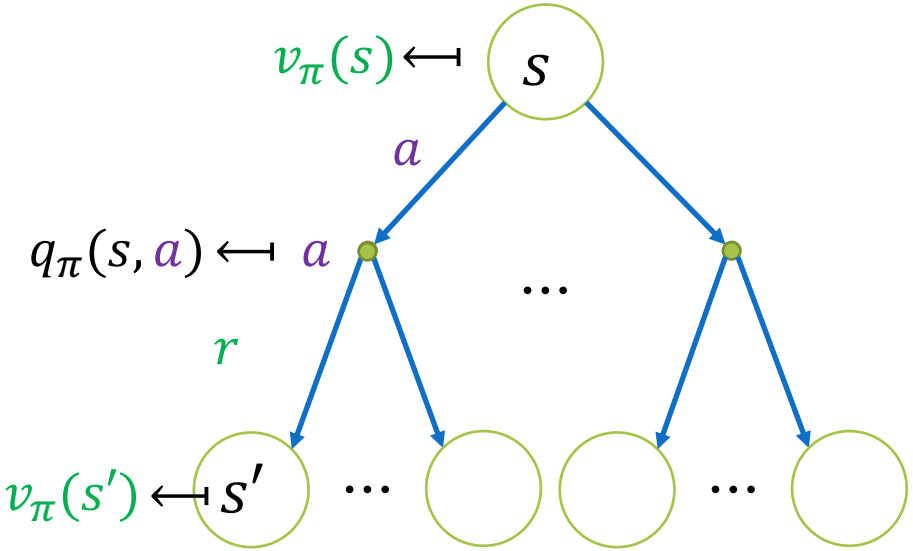
$$v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s)\, q_\pi(s,a)$$

Bellman Expectation Equation for q_π



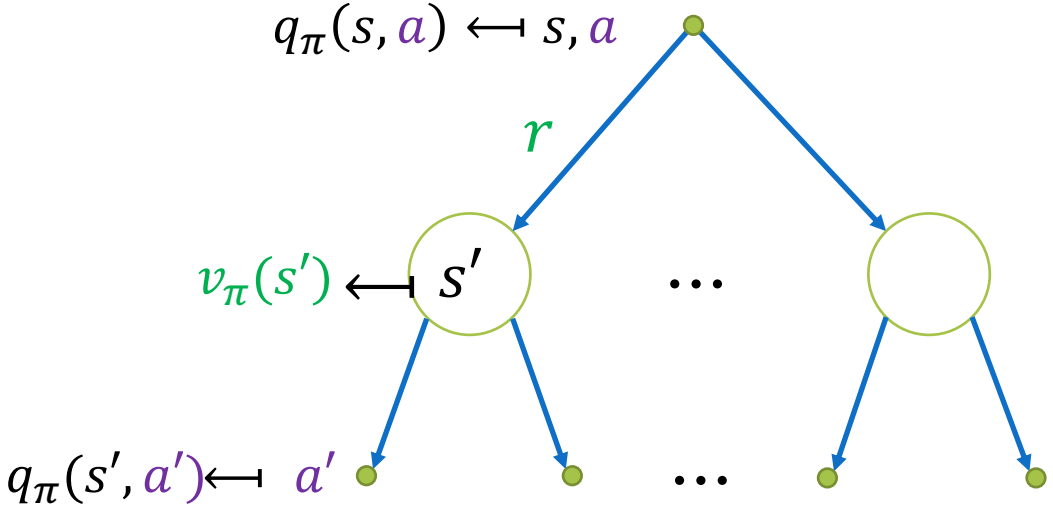
$$q_\pi(s, a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{s,s'}^a \, v_\pi(s')$$

Recursive Formulation for v_π



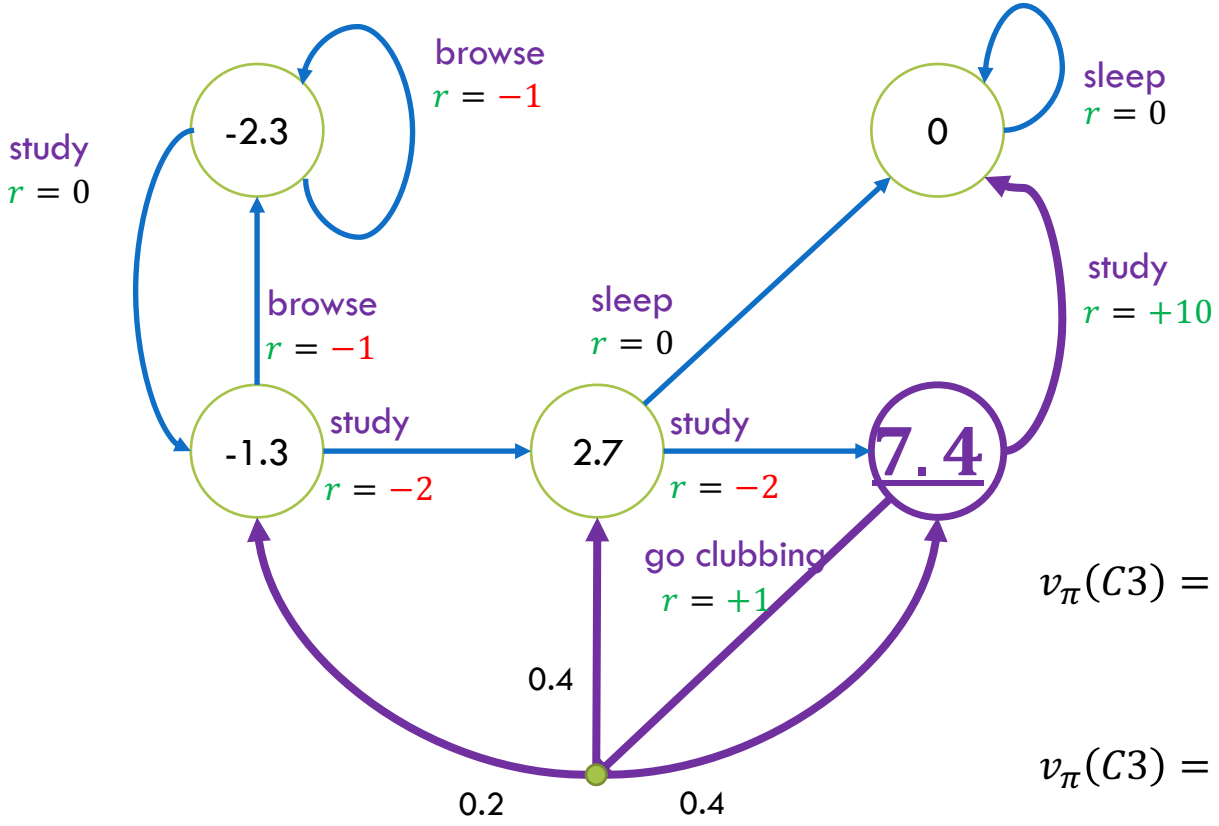
$$v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) \underbrace{\mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{s,s'}^a \, v_\pi(s')}_{q_\pi(s,a)}$$

Recursive Formulation for q_π



$$q_\pi(s,a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{s,s'}^a \underbrace{\sum_{a' \in \mathcal{A}} \pi(a'|s')\, q_\pi(s',a')}_{v_\pi(s')}$$

Bellman Expectation for Student MDP



Under a uniform policy

$$v_\pi(C3) = \frac{1}{2}(10) + \frac{1}{2}(1 + 0.2\times -1.3 + 0.4\times 2.7 + 0.4\times 7.4)$$

$$v_\pi(C3) = 7.4$$

Bellman Expectation Equation (Matrix Form)

The Bellman expectation equation can be expressed concisely using the induced MRP:

$$v_\pi = \mathcal{R}^\pi + \gamma \mathcal{P}^\pi v_\pi$$

with the direct solution:

$$v_\pi = (I - \gamma \mathcal{P}^\pi)^{-1} \mathcal{R}^\pi$$

Optimal Value Functions

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Optimal Value Function

The optimal state-value function $v_*(s)$ is the maximum state–value function over all policies:

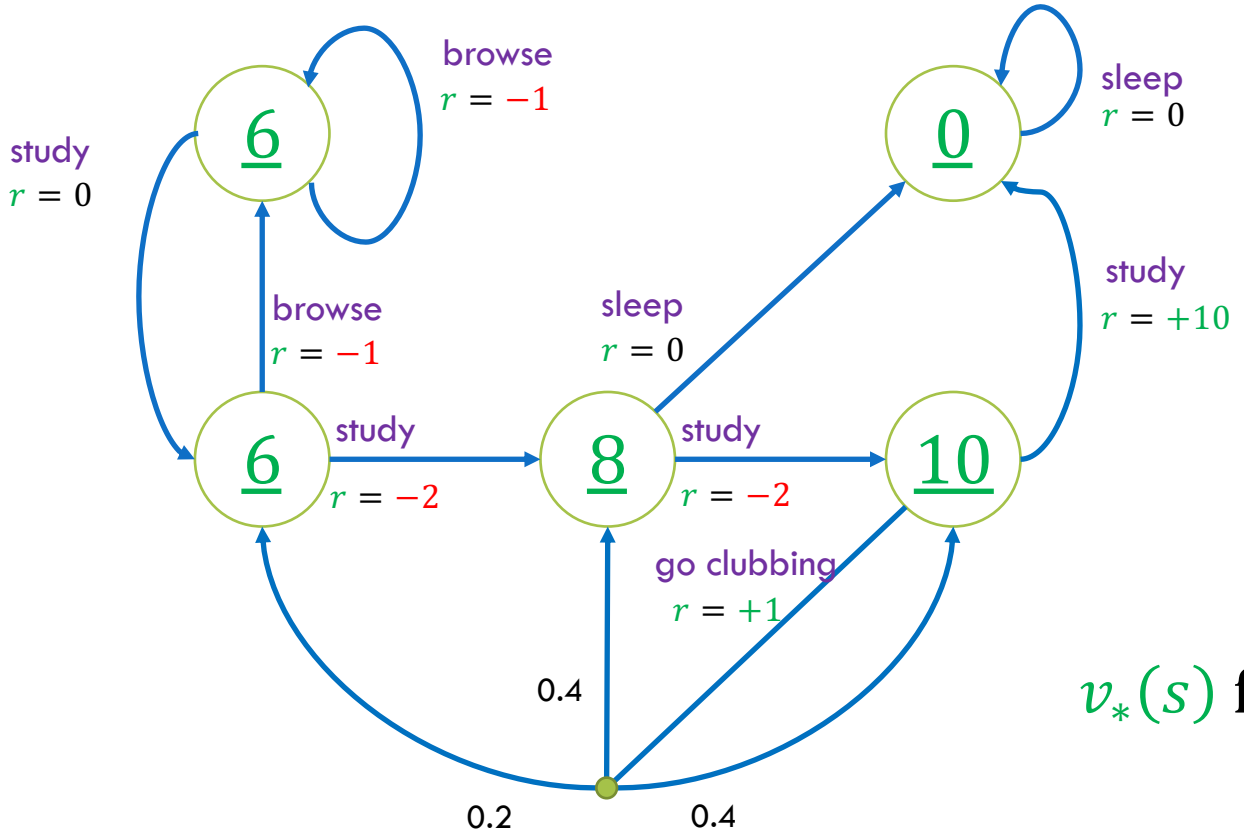
$$v_*(s) = \max_{\pi} v_\pi(s)$$

The optimal action-value function $q_*(s, a)$ is the maximum action–value function over all policies:

$$q_*(s, a) = \max_{\pi} q_\pi(s, a)$$

- These optimal functions specify the best possible performance in the MDP.
- We say that an MDP is solved when we know the optimal value functions.

Optimum Value Function for Student MDP



$v_*(s)$ for $\gamma = 1$

Optimal Policy

Define a partial ordering over the possible policies:

$$\pi \geq \pi' \qquad \text{if} \qquad v_\pi(s) \geq v_{\pi'}(s), \forall s$$

<u>Theorem</u>: For any MDP,

- There exists an optimal policy π_* that is better than or equal to all other policies, $\pi_* \geq \pi, \forall \pi$
- All optimal policies achieve the optimal value function $v_{\pi_*}(s) = v_*(s)$
- They also achieve the optimal action–value function $q_{\pi_*}(s, a) = q_*(s, a)$

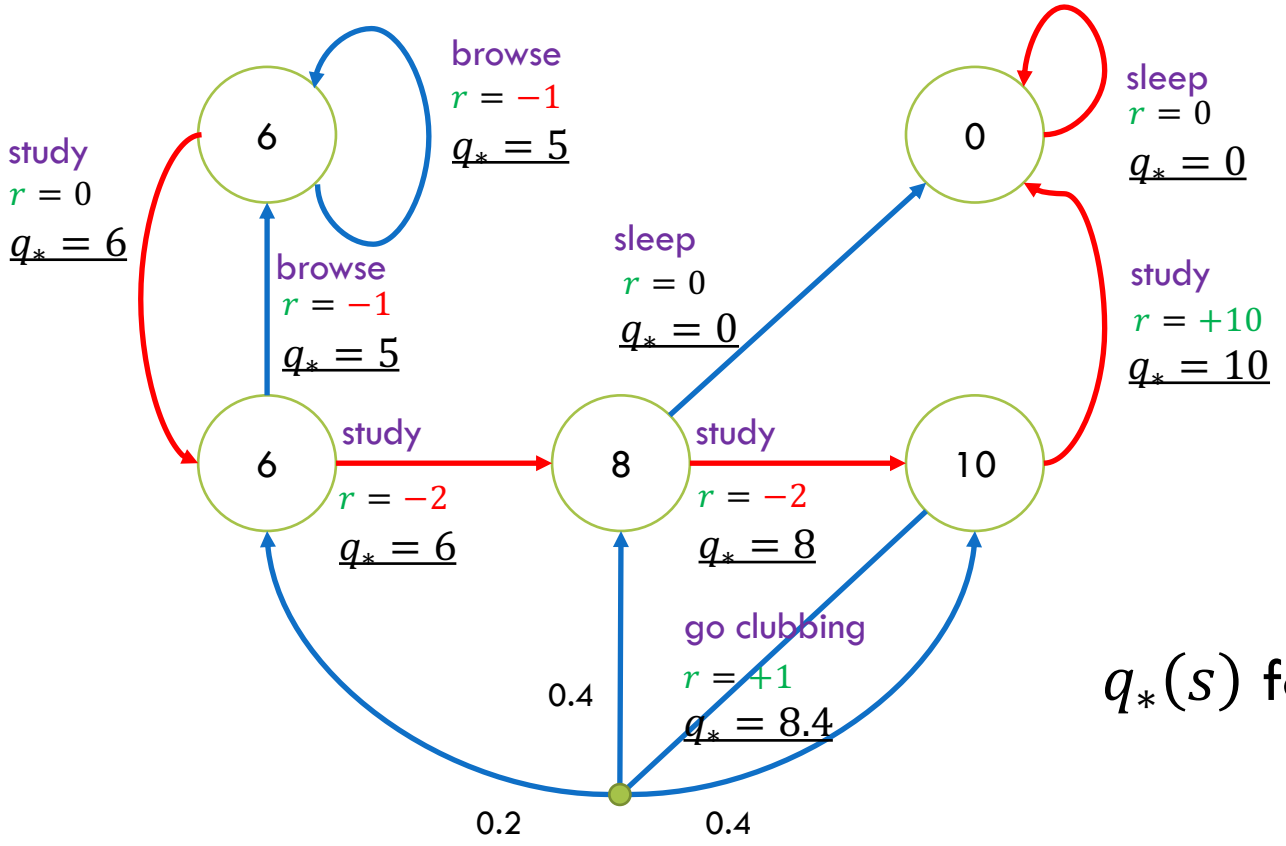
Finding an Optimal Policy

An optimal policy can be found by maximising over $q_*(s, a)$

$$\pi_*(a|s) = \begin{cases} 1 & \text{if } a = \underset{a \in \mathcal{A}}{\operatorname{argmax}}\, q_*(s, a) \\ 0 & \text{otherwise} \end{cases}$$

- There is always a deterministic optimal policy for any MDP
- If we know $q_*(s, a)$, we immediately have the optimal policy

Student Action–Value Function for Student MDP



$q_*(s)$ for $\gamma = 1$

Bellman Optimality

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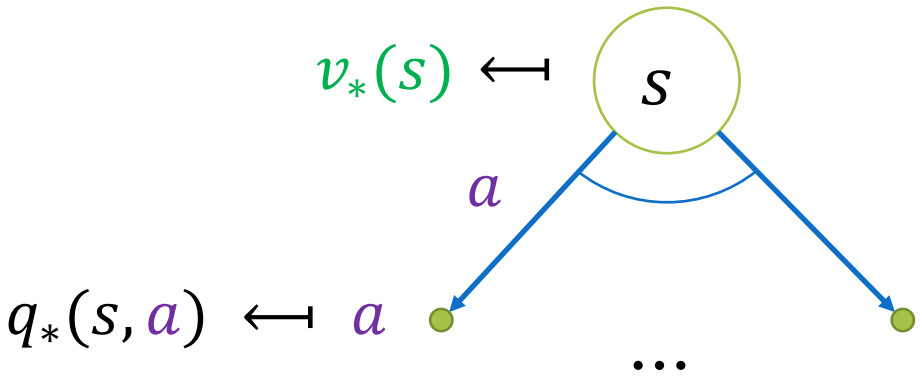
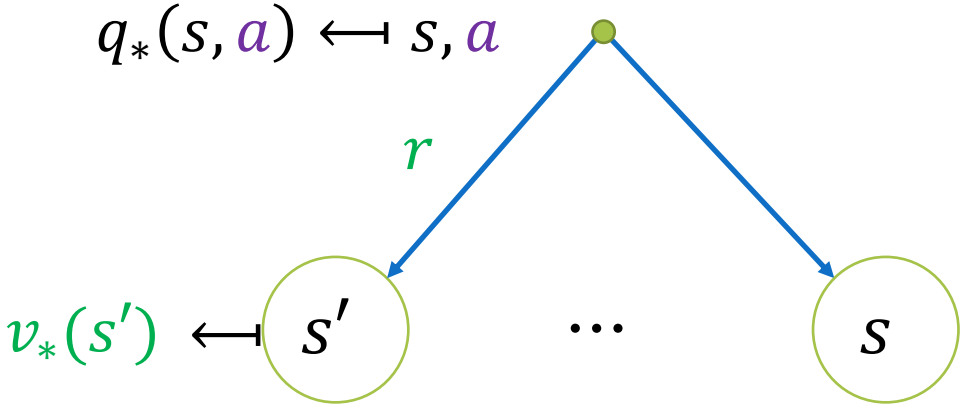
Bellman Optimality Equation for V^*

The optimal value functions are recursively related by the Bellman optimality equations:



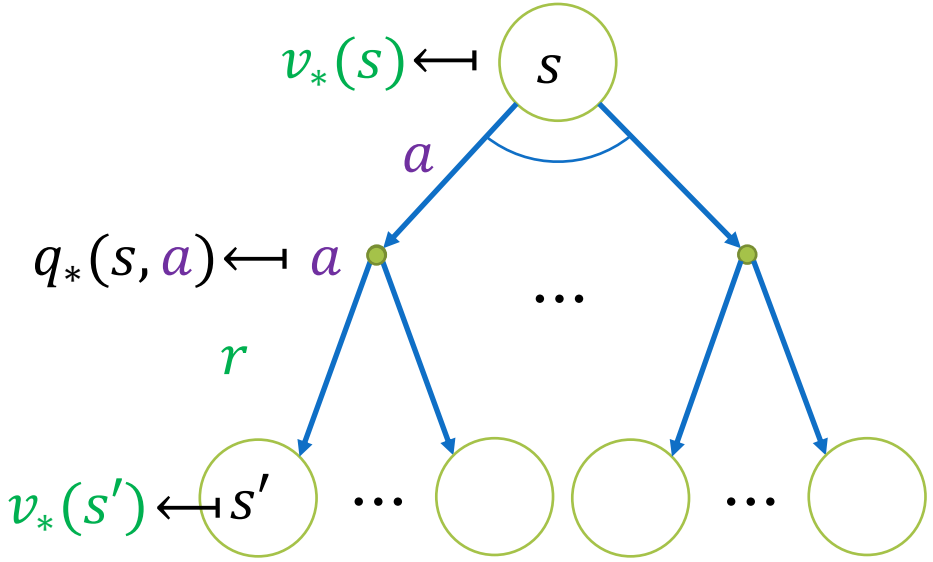
$$v_*(s) = \max_a q_*(s,a)$$

Bellman Optimality Equation for Q^*



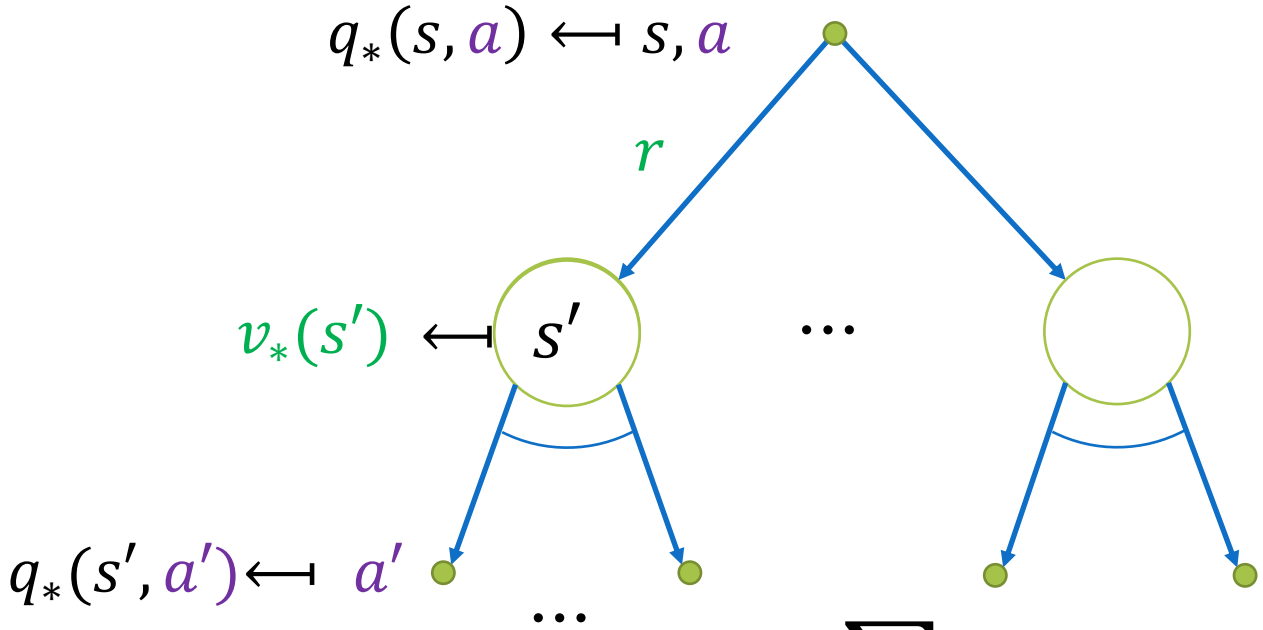
$$q_*(s,a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{s,s'}^a \, v_*(s')$$

Recursive Formulation for V^*



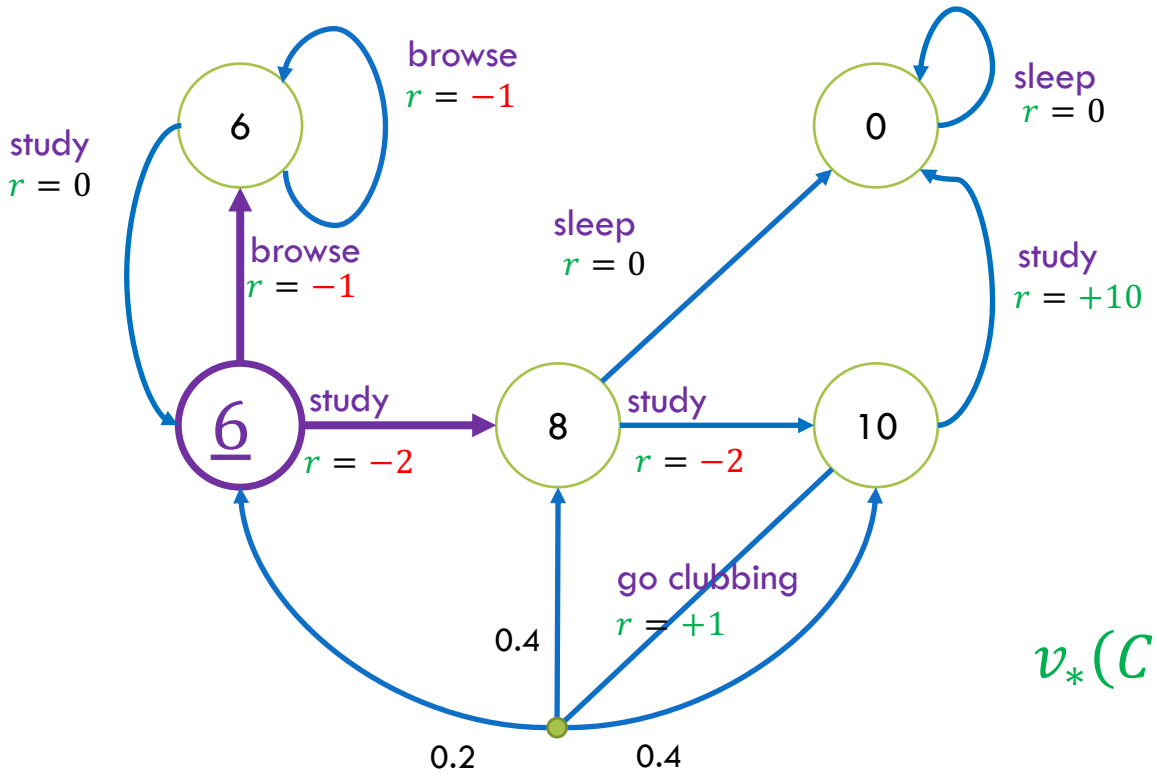
$$v_*(s) = \max_a \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{s,s'}^a \, v_*(s')$$

Recursive Formulation for Q^*



$$q_*(s,a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{s,s'}^a \max_{a'} q_*(s', a')$$

Example Optimality Equation in Student MDP



$$v_*(C1) = \max(-2 + 8, -1 + 6)$$
$$= 6$$

Solving Bellman Optimality Equations

The Bellman Optimality Equation is non-linear

- Why? Due to the $\max$ operations
- No closed form solution (in general)

Many iterative solution methods

- Policy Iteration
- Value Iteration
- Q-learning

Policy and Value Iteration

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Policy Evaluation

Compute state–value function V^{π} for an arbitrary policy π:

$$
\begin{aligned}
V^{\pi}(s) &= \mathbb{E}_{\pi}\{r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \ \ldots | s_t = s\} \\
&= \mathbb{E}_{\pi}\{r_{t+1} + \gamma V^{\pi}(s_{t+1}) | s_t = s\} \\
&= \sum_a \pi(s,a) \sum_{s'} \mathcal{P}^a_{s,s'} \left[\mathcal{R}^a_{s,s'} + \gamma V^{\pi}(s')\right]
\end{aligned}
$$

Why do we want this?

- To evaluate the goodness of a policy

Policy Evaluation

<u>inputs</u>: mdp, an MDP with states $\mathcal{S}$, transitional model $\mathcal{P}$, reward function $\mathcal{R}$, discount γ
π, the policy to be evaluated
<u>local variables</u>: V, a vector of values for states in $\mathcal{S}$, initially zero
Δ, the maximum change in value of any state in an iteration

repeat

$\Delta \leftarrow 0$

for each state $s \in \mathcal{S}$ **:**

$v \leftarrow V(s)$
$V(s) \leftarrow \sum_a \pi(s,a) \sum_{s'} \mathcal{P}^a_{s,s'} \left[\mathcal{R}^a_{s,s'} + \gamma V(s')\right]$
$\Delta \leftarrow \max\left(\Delta, |v - V(s)|\right)$

until $\Delta < \epsilon$

return $V \cong V^\pi$

Policy Improvement

After evaluating a policy, how do we make it better?

What if we take some other action $a \neq \pi(s)$ at state s and the result is better than $V^{\pi}(s)$?

…Then we need to change our policy!

Extend this to consider changes at all states and all possible actions.

Select at each state the action that appears best according to $Q^{\pi}(s, a)$

Policy Iteration

Outer Loop

Inner Loop

1. Initialization
   $V(s) \in \Re$ and $\pi(s) \in \mathcal{A}(s)$ arbitrarily for all $s \in \mathcal{S}$

2. Policy Evaluation
   Repeat
   $\Delta \leftarrow 0$
   For each $s \in \mathcal{S}$:
   $v \leftarrow V(s)$
   $V(s) \leftarrow \sum_{s'} \mathcal{P}_{ss'}^{\pi(s)} \left[\mathcal{R}_{ss'}^{\pi(s)} + \gamma V(s') \right]$
   $\Delta \leftarrow \max(\Delta, |v - V(s)|)$
   until $\Delta < \theta$ (a small positive number)

3. Policy Improvement
   policy-stable $\leftarrow$ *true*
   For each $s \in \mathcal{S}$:
   $b \leftarrow \pi(s)$
   $\pi(s) \leftarrow \arg\max_a \sum_{s'} \mathcal{P}_{ss'}^{a} \left[\mathcal{R}_{ss'}^{a} + \gamma V(s') \right]$
   If $b \neq \pi(s)$, then *policy-stable* $\leftarrow$ *false*
   If *policy-stable*, then stop; else go to 2

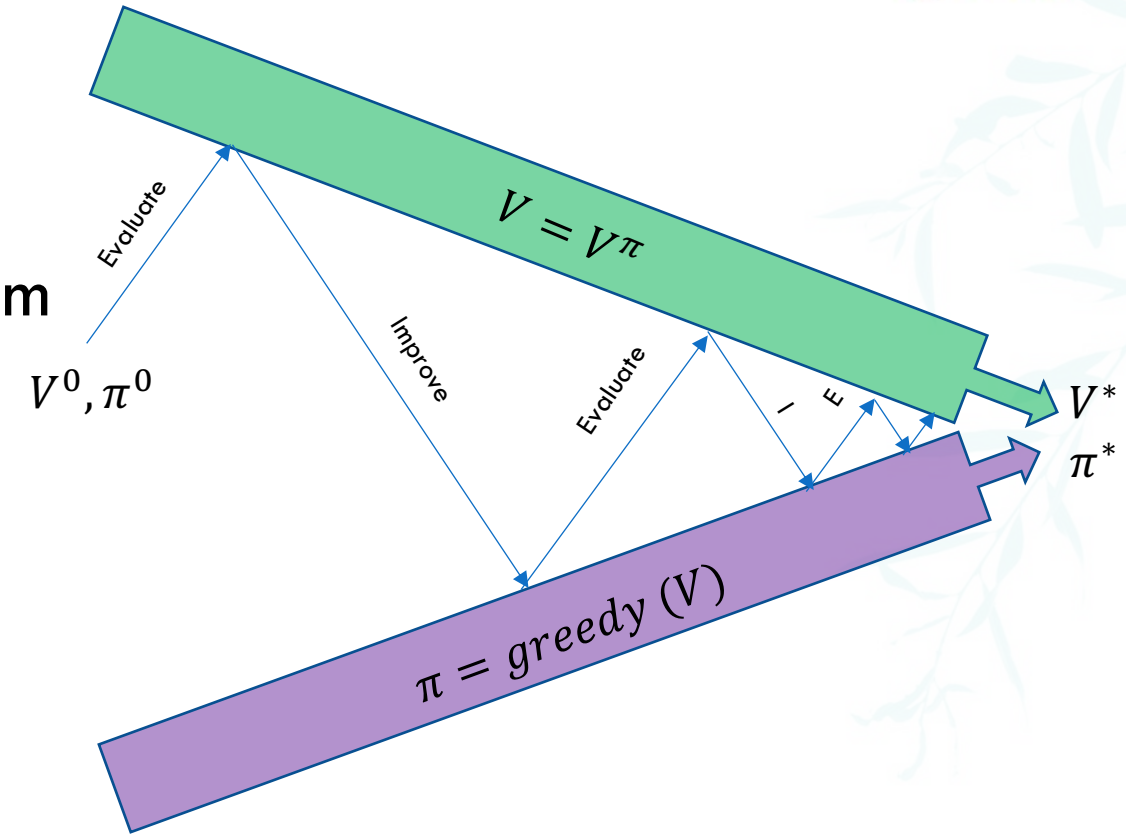
Basic Idea:

Iterate over policies until some stable policy is reached.

Generalised Policy Iteration

Start with a random value function and policy

And iteratively improve them



Our goal: We want π^*.

To achieve this, does policy evaluation need to converge to V^{π}?

Why not update the policy every iteration? → Value iteration.

Value Iteration

In Policy Iteration, each iteration involves a complete policy evaluation, which may itself be a protracted iterative computation, requiring multiple sweeps through the state space.

Instead use Value Iteration: Calculate the return of each state, then use those state returns to find the optimal policy;
i.e., use a truncated form of Policy Evaluation

Value Iteration

<u>inputs</u>: mdp, an MDP with states $\mathcal{S}$, transitional model $\mathcal{P}$, reward function $\mathcal{R}$, discount γ
ϵ, the maximum error allowed in the utility of any state
<u>local variables</u>: V, a vector of values for states in $\mathcal{S}$, initially zero
Δ, the maximum change in value of any state in an iteration

repeat

$\Delta \leftarrow 0$

for each state $s \in \mathcal{S}$ **:**

$v \leftarrow V(s)$
$V(s) \leftarrow \max \sum_{s'} \mathcal{P}^a_{s,s'} \left[\mathcal{R}^a_{s,s'} + \gamma V(s')\right].$
$\Delta \leftarrow \max\left(\Delta, |v - V(s)|\right)$

until $\Delta < \epsilon$

return $V \cong V^*$

… then use V to output the best policy π^*, such that

$$\pi(s) = \arg\max_a \sum_{s'} \mathcal{P}^a_{s,s'} \left[\mathcal{R}^a_{s,s'} + \gamma V(s')\right]$$

Q Learning

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The optimal policy

We want to choose a value-maximizing policy.

However, we are missing two key bits of information:

1. The rewards of unobserved states.
2. What the next state will be when we take an action.

<u>The challenge</u>:
Infer rewards/state transitions as we go along
…while maximizing revenue.

Markov Decision Process

MDP $\equiv \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$

Completely specified by a distribution:

$$\Pr[s_{t+1} = s; r_{t+1} = r | s_t, a_t]$$

"What is the next state and reward given current state and action?"

<u>Planning</u>: given an MDP, compute the optimal policy.

<u>Learning</u>: don't know the MDP, learn a strategy.

Markov Decision Problem

Value / Policy Iteration solves Planning

(we are model based, we know the MDP)

But… in learning, we don't know the underlying MDP.

We need to perform trial-and-error, interact with environment and not lose too much reward along the way.

Q Learning: iteration, redux

Learn this

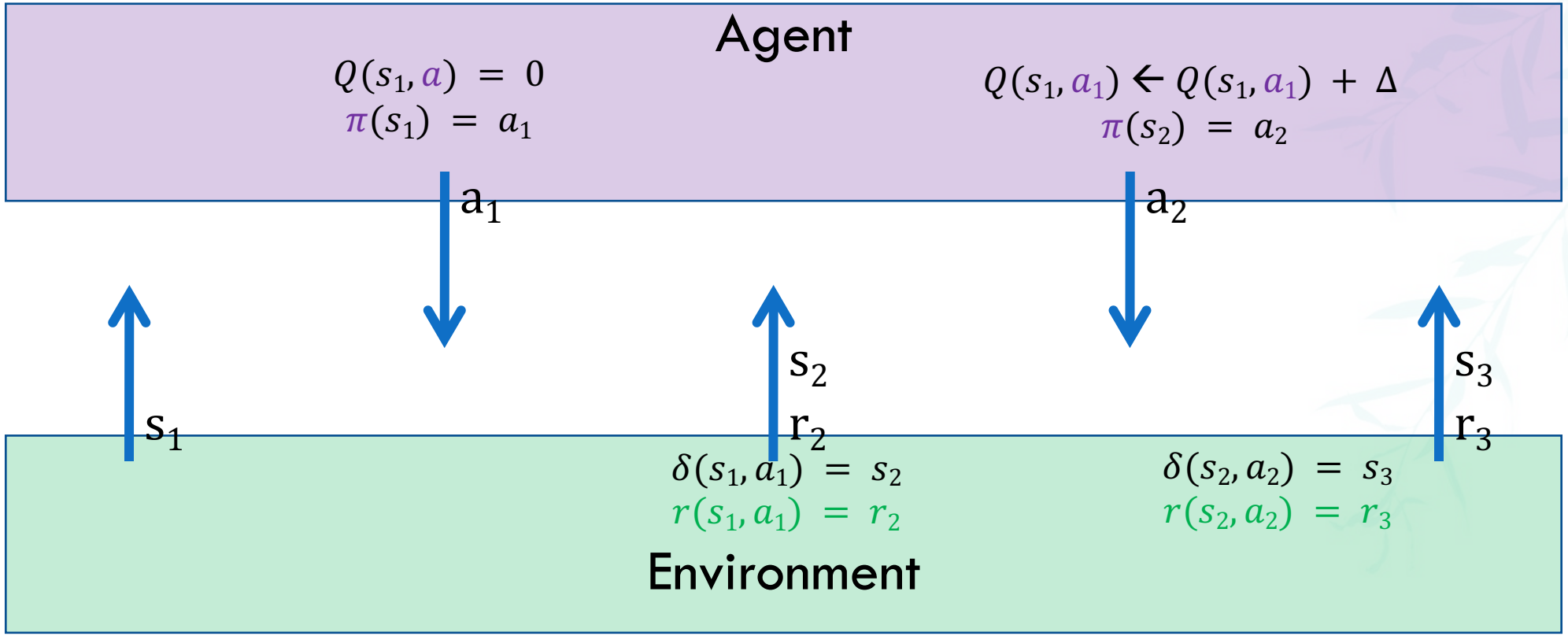
… and this

$$q_*(s,a) = \cancel{\mathcal{R}_s^a} r(s,a) + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{s,s'}^a \max_{a'} q_*(\cancel{s', a'}\ \delta(s,a))$$

We don't have the optimal Q function Q_* at first …

- Let's act as if we do
- Updates will bring it closer to optimal … until it is optimal!

Q-Learning: The Procedure



Q Learning

<u>local variables</u>: $\hat{Q}$, a vector of Q values for state action pairs in $\mathcal{S} \times \mathcal{A}$, initially zero

$s \leftarrow s_0$

repeat

select action a

receive reward $r(s,a)$

$s' \leftarrow \delta(s,a)$

$\hat{Q}(s,a) \leftarrow r(s,a) + \gamma \max \hat{Q}(s',a')$

Key observation: When $r(s,a) \geq 0$ and $\hat{Q} = 0, \hat{Q}(s,a) \leq Q^*(s,a)$ always.

In other words – we always underestimate the optimal Q values.

They always increase at every iteration, thus we converge to Q^*… and in particular to an optimal policy!

Handling Non-determinism

local variables: $\hat{Q}$, a vector of Q values for state action pairs in $\mathcal{S}\times\mathcal{A}$, initially zero

$s \leftarrow s_0$
repeat
select action a
receive reward $r(s,a)$
$s' \leftarrow \delta(s,a)$

$$\hat{Q}(s,a) \leftarrow r(s,a) + \gamma \max \hat{Q}(s',a')$$

Problem 1: both $r(s,a)$ and $r(s,a)$ are stochastic. We should factor current $\hat{Q}(s,a)$ in the computation instead of replacing it entirely.

Solution: update $\hat{Q}(s,a)$ in a gradual manner, as we observe it more:

$$\hat{Q}(s,a) \leftarrow (1-\alpha)\cdot\left[\hat{Q}(s,a)\right] + \alpha \cdot \left[r(s,a) + \gamma \max \hat{Q}(s',a')\right], \text{ where } \alpha = \frac{1}{1+N[s,a]}$$

# of times a taken at state s

Exploration vs. Exploitation

local variables: $\hat{Q}$, a vector of Q values for state action pairs in $\mathcal{S} \times \mathcal{A}$, initially zero

$s \leftarrow s_0$
repeat

select action a
receive reward $r(s, a)$
$s' \leftarrow \delta(s, a)$
$\hat{Q}(s, a) \leftarrow r(s, a) + \gamma \max \hat{Q}(s', a')$

Problem 2: How do we select an action a? Greedily? Then we could miss out better actions.

First Cut Solution: pick action a greedy, but at random some small percentage of time (a.k.a. ϵ-greedy)

Your Turn: Can you suggest a better method for choosing action a?

Summing Up: Planning versus Learning

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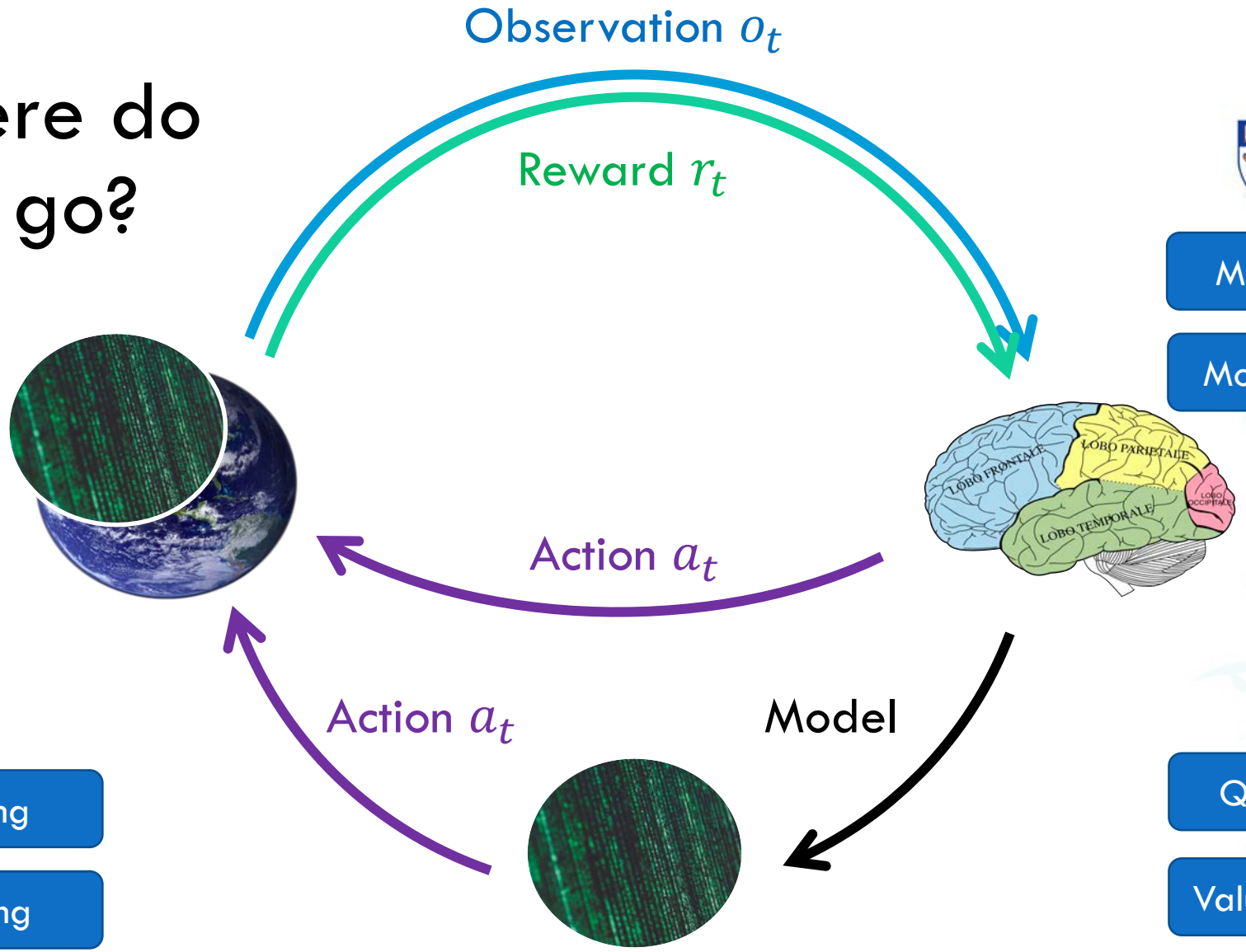
Where do they go?



Model Free

Model Based

Q Learning

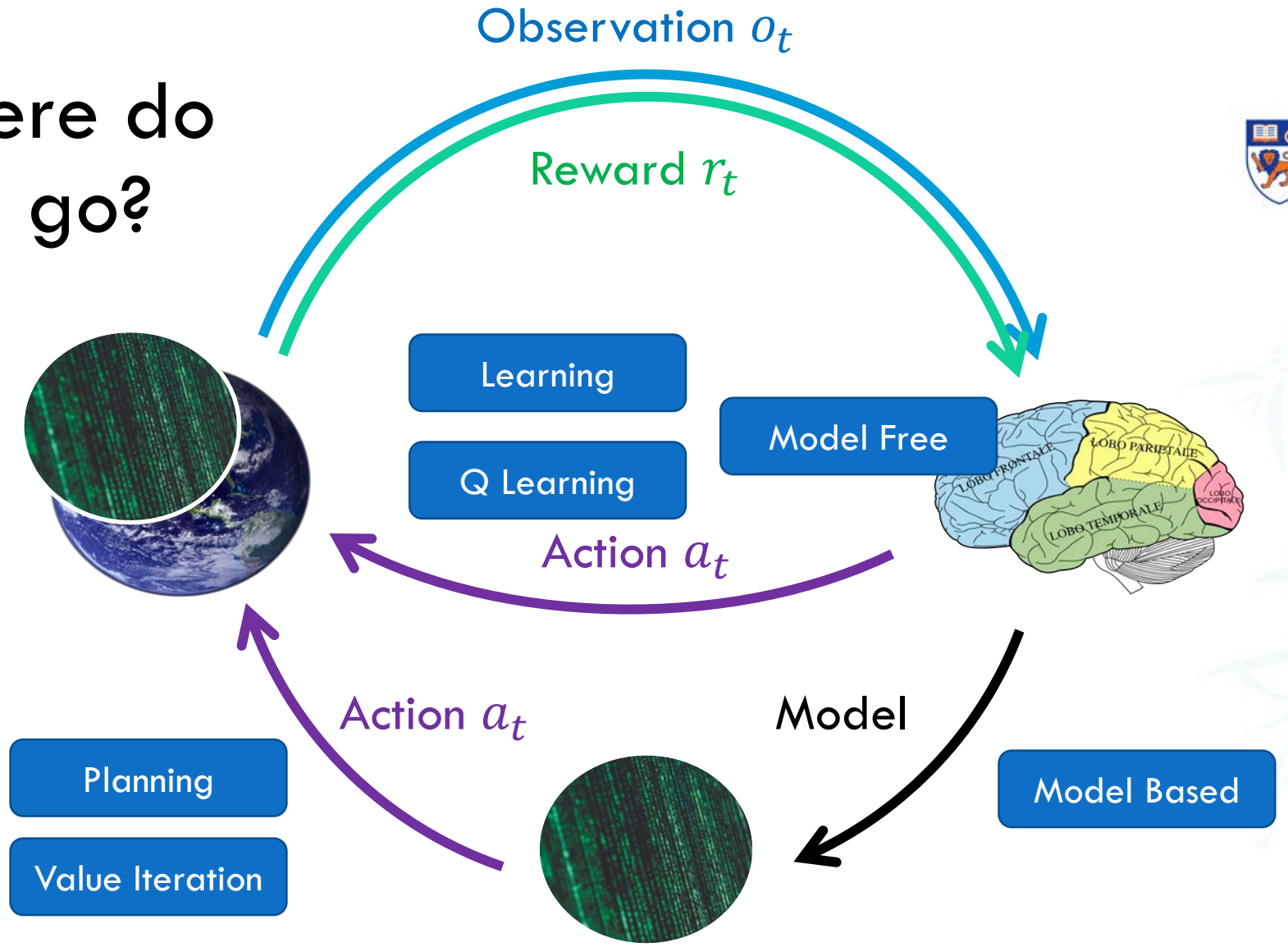
Value Iteration

Planning

Learning

Action a_t

Where do they go?



Action a_t